Figure 32.6 The Watch window of Visual Studio showing the result of validation using the custom `ValidationAttribute`. The user has provided an invalid `currencyTo` value, `£$%`. Consequently, `ModelState` isn't valid and contains a single error with the message `"Not a valid currency code"`.

The main feature missing from this custom attribute is client-side validation. You've seen that the attribute works well on the server side, but if the user entered an invalid value, they wouldn't be informed until after the invalid value had been sent to the server. That's safe, and it's as much as you need to do for security and data-consistency purposes, but client-side validation can improve the user experience by providing immediate feedback.

You can implement client-side validation in several ways, but it's heavily dependent on the JavaScript libraries you use to provide the functionality. Currently, ASP.NET Core Razor templates rely on jQuery for client-side validation. See the "Custom client-side validation" section of Microsoft's "Model validation in ASP.NET Core MVC and Razor Pages" documentation for an example of creating a jQuery Validation adapter for your attributes: http://mng.bz/Wd6g.

TIP Instead of using the official jQuery-based validation libraries, you could use the open source aspnet-client-validation library (https://github.com/haacked/aspnet-client-validation) as I describe on my blog at http://mng.bz/AoXe.

Another improvement to your custom validation attribute would be to load the list of currencies from a DI service, such as an `ICurrencyProvider`. Unfortunately, you can't use constructor DI in your `CurrencyCodeAttribute`, as you can pass only constant values to the constructor of an `Attribute` in .NET. In chapter 22 we